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Simplification

IMPORTANT FACTS AND FORMULAE

I. 'BODMAS' Rule: This rule depicts the correct sequence in which the operations are to be executed, so as to find out the value of a given expression.

Here, 'B' stands for 'Bracket', 'O' for 'of', 'D' for 'Division', 'M' for 'Multiplication', 'A' for 'Addition' and 'S' for 'Subtraction'.

Thus, in simplifying an expression, first of all the brackets must be removed, strictly in the order (), { } and [].

After removing the brackets, we must use the following operations strictly in the order:

(i) of (ii) Division (iii) Multiplication (iv) Addition (v) Subtraction

II. Modulus of a Real Number: Modulus of a real number a is defined as

$$|a| = \begin{cases} a, & \text{if } a > 0 \\ -a, & \text{if } a < 0 \end{cases}$$

Thus, $|5| = 5$ and $|-5| = -(-5) = 5$.

III. Virnaculum (or Bar): When an expression contains Virnaculum, before applying the 'BODMAS' rule, we simplify the expression under the Virnaculum.

IV. Some Important Formulae:

$$(i) (a + b)^2 = (a^2 + b^2 + 2ab)$$

$$(ii) (a - b)^2 = (a^2 + b^2 - 2ab)$$

$$(iii) (a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$$

$$(iv) (a + b)^2 - (a - b)^2 = 4ab$$

$$(v) (a^2 - b^2) = (a + b)(a - b)$$

$$(vi) (a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$(vii) (a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

$$(viii) (a^3 + b^3) = (a + b)(a^2 - ab + b^2)$$

$$(ix) (a^3 - b^3) = (a - b)(a^2 + ab + b^2)$$

$$(x) (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$(xi) a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

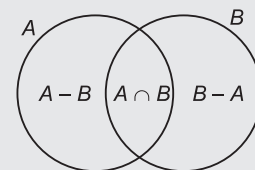
V. For any two sets A and B, we have:

$$(i) n(A - B) + n(A \cap B) = n(A)$$

$$(ii) n(B - A) + n(A \cap B) = n(B)$$

$$(iii) n(A \cup B) = n(A - B) + n(A \cap B) + n(B - A)$$

$$(iv) n(A \cup B) = n(A) + n(B) - n(A \cap B).$$



SOLVED EXAMPLES

Ex. 1. $4368 + 2158 - 596 - ? = 3421 + 1262$

(Bank P.O., 2009)

Sol. Let $4368 + 2158 - 596 - x = 3421 + 1262$

$$\Rightarrow x + 596 = (4368 + 2158) - (3421 + 1262)$$

$$\Rightarrow x + 596 = 6526 - 4683 = 1843$$

$$\Rightarrow x = 1843 - 596 = 1247.$$

Hence, required number = 1247.

Ex. 2. $3456 \div 12 \div 8 = ?$

(Bank Recruitment, 2008)

Sol. Given expression = $\frac{3456}{12} \div 8 = 288 \div 8 = 36$.